National
Qualifications
2019

2019 Computing Science

Higher

Finalised Marking Instructions

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These marking instructions have been prepared by examination teams for use by SQA appointed markers when marking external course assessments.

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General marking principles for Higher Computing Science

Always apply these general principles. Use them in conjunction with the detailed marking instructions, which identify the key features required in candidates' responses.

(a) Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted for errors or omissions.

(b) If a candidate response does not seem to be covered by either the principles or detailed marking instructions, and you are uncertain how to assess it, you must seek guidance from your team leader.

(c) Award marks regardless of spelling, as long as the meaning is unambiguous.

(d) Candidates may answer programming questions in any appropriate programming language or pseudocode. Award marks regardless of minor syntax errors, as long as the intention of the coding is clear.

(e) For a **describe** question, candidates must provide a statement or structure of characteristics and/or features. This will be more than an outline or a list. It may refer to, for example, a concept, process, experiment, situation, or facts, in the context of and appropriate to the question. Candidates must make the same number of factual/appropriate points as there are marks available in the question.

(f) For an **explain** question, candidates must relate cause and effect and/or make relationships between things clear, in the context of the question or a specific area within the question.

Marking instructions for each question

SECTION 1

<table>
<tr><th colspan="3">Question</th><th>Expected response</th><th>Max mark</th><th>Additional guidance</th></tr>
<tr><td>1.</td><td></td><td></td><td>-102</td><td>1</td><td></td></tr>
<tr><td>2.</td><td></td><td></td><td>• Both substrings
• Assignment and concatenation</td><td>2</td><td>1 mark for each bullet
Examples:
shortDate = left(month,3) & right(year,2)

shortDate = mid(month, 1, 3) & mid(year,3,2)

shortDate = month[0:3] + year[2:4]
shortDate = month[:3]+ year[2:] OR year[-2:]

shortDate = month[1]+ month[2]+ month[3] + year[3]+year[4]</td></tr>
<tr><td>3.</td><td></td><td></td><td>• The processor activates the read line on the control bus/the control unit activates the read line.
• An instruction is fetched from the memory location using the data bus (and stored in the instruction register).</td><td>2</td><td>1 mark for each bullet

Read line and control bus/unit required

From memory/to processor and data bus required</td></tr>
<tr><td>4.</td><td></td><td></td><td>• Sign bit: 0
• Remaining 15 bits of mantissa:
110 0010 0000 0000
• Exponent: 0000 0011</td><td>3</td><td>1 mark for each bullet

(9 x 0s)
(6 x 0s)</td></tr>
<tr><td>5.</td><td></td><td></td><td>The client:
• Details requirements
• Outlines scope and boundaries
• Evaluates prototype/suggest changes
• Provides feedback/liaising with development team throughout process</td><td>2</td><td>1 mark for each bullet
Maximum 2 marks</td></tr>
<tr><td>6.</td><td></td><td></td><td><table><tr><th>course</th><th>least expensive item</th></tr><tr><td>Starter</td><td>2.99</td></tr><tr><td>Dessert</td><td>8.99</td></tr></table></td><td>2</td><td>1 mark for each correct row
Maximum 2 marks

1 mark if two correct, one wrong</td></tr>
<tr><td>7.</td><td></td><td></td><td>A key comprising two or more attributes which are foreign/primary keys from other tables</td><td>1</td><td></td></tr>
</table>

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Question			Expected response	Max mark	Additional guidance
8.	(a)		• Two drop down boxes have no indication of the inputs • No context for media elements • Inconsistent image alignment or size under Book Now/bottom right • File formats of the media used are not specified • No URLs specified for navigation links	**2**	**1 mark** for each bullet Maximum 2 marks
	(b)		• Provide scenario/give prospective users a task to perform • Observe users • Receive feedback • Identify problems with navigation eg missing/orphan pages/ hyperlink destinations • Record difficulties or improvements needed	**2**	**1 mark** for each bullet Maximum 2 marks No mark for testing hyperlinks without reference to destination
9.			A – Nav B - Footer	**2**	**1 mark** for each line Accept with < >
10.	(a)		2	**1**	
	(b)		• Use a conditional loop • With termination	**2**	**1 mark** for each bullet Termination examples include target is found, target=list[index], found =true
	(c)	(i)	0	**1**	
		(ii)	• Logic • Returns a position for an item that is not in the list OR Returns the position of the first item in the list/C232	**2**	**1 mark** for each bullet

SECTION 2

Question			Expected response	Max mark	Additional guidance
11.	(a)		• 200 readings/values • Accelerator OR brake maximum 100 (%) OR minimum 0 • Speed minimum is 0 OR maximum dependent on car • Data is accurate as it is captured from sensors. • No input validation required	**2**	**1 mark** for each bullet Maximum 2 marks
	(b)	(i)	• Record structure with name • Four fields specified	**2**	**1 mark** for each bullet Record structure could be Type, Structure, Class, Named Tuple, Record name Is Livecode Global CarData Put empty into CarData[1][speed] Put empty into CarData[1][accel] Put empty into CarData[1][brake] Put empty into CarData[1][seatb] Defining a RECORD structure does not start with DECLARE – do not award first bullet
		(ii)	• Array structure • Using data type created in (i)	**2**	**1 mark** for each bullet Example Python: readings=[datatype_a() for x in range (200)] Example VB: DIM readings(199) AS datatype_a Example Livecode: Set the data of EventData to CarData
	(c)		• Initial set of max • Loop traversing array with termination • If condition • Assignment of max	**4**	**1 mark** for each bullet Set max to first speed Loop to end of array If current speed > max then Set max to current speed (End If End Loop) Do not penalise incorrect shapes in graphical notations.

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Question			Expected response	Max mark	Additional guidance

12.	(a)		• grouping selector for header, main elements with properties
			• additional header selector with height property
				2	**1 mark** for each bullet
					
					`header, main {`
					`  background-color:White;`
					`  margin-top:20px;`
					`  margin-bottom:20px}`
					
					Allow `margin:20 0 20 0;`
					
					`header {`
					`  height:90px}`
					
					header, main selector must be separated with comma

	(b)		• Suitable control for session (radio/list) • Name, Mobile, Date, Additional Information inputs • Validation of any one input from bullet 2 • Submit button	**4**	**1 mark** for each bullet Date picker/calendar/formatted date validates date input
	(c)		Presence check	**1**	
	(d)	(i)	*<input type ="number" name ="days" min ="1" max ="7">*	**1**	Accept min and max without " "
		(ii)	Change the maxlength to a more appropriate length OR example: `maxlength="40"` OR Change the maxlength to minlength OR Remove maxlength	**1**	
	(e)		• Reduced screen area (required for large number of options) • Multiple selections can be made	**2**	**1 mark** for each bullet
	(f)		• Browser/device problems	**1**	

Question			Expected response	Max mark	Additional guidance
13.	(a)		• (A query to) display the total donations made to each fundraiser • (A query to) allow the charity to display total donations made • (A query to) display the total donations from a sponsor(s) • (A query to) display the total donations from a fundraiser(s) • (A query to) insert new fundraisers/sponsors or make a new donation	**2**	**1 mark** for each bullet Maximum 2 marks
	(b)			**2**	

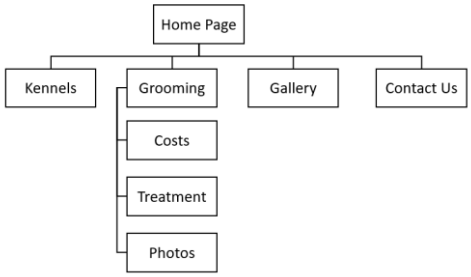
	(c)		• Select clause with fundraiserID field and average aggregate function with amount field
			• Alias (allow missing £ sign) and From
			• Group by fundraiserID
				3	**1 mark** for each bullet
					
					Example:
					
					`SELECT fundraiserID,`
					`  AVG(amount)`
					`As 'Average Donation (£)'`
					`FROM Donation`
					`GROUP BY fundraiserID`
					
					Do not penalise use of extra table

	(d)		• Aggregate function should be MAX(amount) • No alias implemented (Largest donation (£)) • Should have a Group By (Fundraiser.)name	**3**	**1 mark** for each bullet
	(e)		• Public key encrypts the data • Private key decrypts the data	**2**	**1 mark** for each bullet 1 mark for the concept of encrypt and decrypt without assigning to appropriate keys

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Question			Expected response	Max mark	Additional guidance
14.	(a)		• In: stars[], price[] • Out: position	**2**	**1 mark** for each bullet Any additional entries in a row, award 0 marks for that row.
	(b)		• Initialisation of position and lowest that does not cause an error ie position < 0 and lowest as a suitable high value • Condition stars = 5 • Condition price < lowest value • Assign lowest OR use of price (position) in the condition • Assign position	**5**	**1 mark** for each bullet Set position to -1 Set lowest to 1000 Loop for each hotel If current stars = 5 and current price < lowest Set lowest to current price Set position to index End if End loop Award 4 marks if position of price is used eg Set position to -1/0 (no mark) Loop for each hotel If current stars=5 and current price<price [position] Set position to current index End if End loop Do not penalise incorrect shapes in graphical notations.
15.	(a)		• Three arrays of 200 elements • Data types of each array: string, string, integer	**2**	**1 mark** for each bullet
	(b)		• initialise and increment count • loop with termination • condition matching category • condition matching score >= qualScore • output statement concatenating count, message and qualScore	**5**	**1 mark** for each bullet DECLARE count INITIALLY O RECEIVE qualScore FROM KEYBOARD FOR index FROM 0 to 199 DO If category[index] =“Junior” AND score[index]>= qualScore THEN SET count to count +1 END IF NEXT FOR SEND count & “Juniors achieved the qualifying score of ”& qualScore Concatenation can be &, +, comma

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Question			Expected response	Max mark	Additional guidance
16.	(a)		• 4 main pages in top level • 3 sub-pages from Grooming	**2**	**1 mark** for each bullet 

	(b)		• background-color in header selector OR border-color
			• margin in imageBanner class
			• float:right in h1 selector
				3	**1 mark** per bullet
					`header{`
					` **background-color/border-**`
					` **color**:lightgrey;`
					` padding:0px;`
					` }`
					`.imageBanner{`
					` **margin**:20px;`
					` float:left;`
					` padding:0px;`
					` height:90px;`
					` width:130px`
					` }`
					`h1{`
					` **float:right;**`
					` margin-right:30px;`
					` margin-top:30px;`
					` color:white;`
					` display:inline;`
					` }`

	(c)	(i)	• style will only apply within the (parent) nav element • style will apply to unordered list/list and/or anchor elements	**2**	**1 mark** for each bullet Exemplification of bullet two is acceptable
		(ii)	The background colour changes to black and the text colour changes to white	**1**	
		(iii)	display:block	**1**	

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Question			Expected response	Max mark	Additional guidance
16.	(d)	(i)	• Appropriate function name with parameter in brackets • Change source attribute of the parameter to Bailey1 image	**2**	**1 mark** for each bullet `function mouseaway(my_image) {my_image.src = "../images/Bailey1.png";}` Accepts solution using document.getElementByID() Do not penalise { }, ” ”, ; Ensure new function is not named rollover
		(ii)	• Add onmouseout event to `<img>` element • Match function name to name created in part (i)	**2**	**1 mark** for each bullet `<img id='dog1Image' src = "../images/Bailey1.png" onmouseover="rollover(this)" onmouseout="mouseaway(this)">`
	(e)		Pro-active decision making by software/apps to reduce usage eg • automatically switch heating off/on when within range • monitoring/responding to external weather conditions • adjust heating based on historical patterns of data • zoned heating	**1**	**1 mark** for any one bullet

Question			Expected response(s)	Max mark	Additional guidance
17.	(a)		• forename, surname, SUM(lessonDuration) • selection of date in search • grouping by forename and surname	**3**	**1 mark** for each bullet Date selection may be between, range and use wildcards (* ?? %)
	(b)		• SUM aggregate function • Calculation of lessonduration * 35 • Alias and Booking table	**3**	**1 mark** for each bullet Example: `SELECT SUM(lessonDuration*35) AS 'Total Payments' FROM Booking;` Note can be (lessonDuration) * 35
	(c)		• UPDATE Customer table • SET new address and postcode values • WHERE criteria based on forename and surname OR customer ID	**3**	**1 mark** for each bullet Example: `UPDATE Customer SET Address = '27 Drummer Street', Postcode = 'AB33 7QR' WHERE forename = 'Simon' AND surname = 'Wenger'` OR `customerID = 2`
	(d)		• forename, surname and COUNT function on any field • Tables(s): Instructor and Booking • Grouping: forename and surname • Sort Order: Count of field descending	**4**	**1 mark** for each bullet COUNT function can be applied to any field or * Accept ALIAS in the Sort Order

<table>
<tr><th colspan="3">Question</th><th>Expected response(s)</th><th>Max mark</th><th>Additional guidance</th></tr>
<tr><td>18.</td><td>(a)</td><td>(i)</td><td>List or newList</td><td>1</td><td></td></tr>
<tr><td></td><td></td><td>(ii)</td><td>Emails or uniques</td><td>1</td><td></td></tr>
<tr><td></td><td>(b)</td><td>(i)</td><td>Section of code in which a variable is accessible/usable</td><td>1</td><td></td></tr>
<tr><td></td><td></td><td>(ii)</td><td>Global or a description of global</td><td>1</td><td>Lines 1 to 79</td></tr>
<tr><td></td><td>(c)</td><td>(i)</td><td><table><tr><th>Variable</th><th>Value</th></tr><tr><td><code>position</code></td><td>1</td></tr><tr><td><code>newList[position]</code></td><td>bert@yeeha.com</td></tr><tr><td><code>list[index]</code></td><td>bert@yeeha.com</td></tr></table></td><td>3</td><td>1 mark for each row

Award 2 marks for:
position 2
newList[position]: colin@iclood.com
list[index]: colin@iclood.com

Award 1 mark for:
position 0
newList[position]: allan@giggle.com</td></tr>
<tr><td></td><td></td><td>(ii)</td><td>• bert@yeeha.com or list(1) does not equal allan@giggle.com or newlist(0) therefore condition is true
• newList(1) is assigned the value bert@yeeha.com (list(1))</td><td>2</td><td>1 mark for each bullet

Accept answers that compare values from trace table in (c)(i)

Award 1 mark for meaningful description that does not reference given values in the array</td></tr>
<tr><td></td><td>(d)</td><td></td><td>• Slow performance
• Inability to access</td><td>1</td><td>1 mark for each bullet
Maximum 1 mark</td></tr>
</table>

[END OF MARKING INSTRUCTIONS]

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